

# Maven AI Investment Copilot – Setup & Run Guide

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## Prerequisites

- Node.js 18+
- Python 3.10+
- Git
- A Google Gemini API Key

## Project Structure

frontend/ -> Next.js UI & Node.js API Gateway

ai-service/ -> Python AI Microservice

## Environment Variables

Create a .env file in the ai-service directory before running the project.

### AI Service (.env)

```
ENV=development
HOST=127.0.0.1
PORT=8000
GEMINI_API_KEY=your_api_key_here
GEMINI_MODEL=gemini-2.5-flash
```

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## Installation

### 1. Clone Repository

```
git clone <repository-url>
cd Maven
```

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### 2. Install Frontend & API Gateway Dependencies

```
cd frontend
npm install
```

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### 3. Install AI Service Dependencies

```
cd ai-service
python -m venv .venv

# Linux/macOS:
source .venv/bin/activate
# Windows:
.venv\Scripts\activate

pip install -r requirements.txt
```

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## Running the Application

### Terminal 1 (Start the AI Service)

```
cd ai-service
# Ensure virtual environment is activated
python -m uvicorn src.main:app
```

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### Terminal 2 (Start the Frontend & API Gateway)

```
cd frontend
npm run dev
```

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## Access the Application

- Frontend UI: <http://localhost:3000>
- Backend API Gateway: <http://localhost:3000/api/v1>
- AI Service: <http://localhost:8000>

## Build for Production

### Frontend & API Gateway

```
cd frontend
npm run build
npm start
```

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## AI Service

```
cd ai-service  
python -m uvicorn src.main:app
```

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## Common Issues

- Verify that the Gemini API key is configured correctly in ai-service/.env.
- Ensure the AI Service (Terminal 1) is running before submitting a search in the Frontend.
- Check that the services are running on their default ports (3000 and 8000).